

Vivliostyle Themes Documentation

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Vivliostyle Themes



Vivliostyle Themes logo

Vivliostyle Theme is a style theme used for creating publications with Vivliostyle. With Vivliostyle Themes, you can effortlessly change the style of your publications.

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Adoption of the Official Theme

A generic theme that is likely to be used by many users will be adopted as the official theme. Official themes are managed in the `vivliostyle/themes` repository (<https://github.com/vivliostyle/themes>) and published as npm packages with the name `@vivliostyle/theme-*`.

Conditions for Adoption

The conditions for adoption as an official theme are as follows:

- The design and primary purpose of the theme must be different from the existing official themes.
- The theme must meet the `theme specifications`.
- Sign the Contributor License Agreement.
- Comply with the `Code of Conduct`.

How to Adopt

Please send a pull request to the `vivliostyle/themes` repository (<https://github.com/vivliostyle/themes>).

- The theme name should be `@vivliostyle/theme-*`.
- Include the source code in `packages/@vivliostyle/theme-*`.

If your theme is accepted as an official theme, you will be given the following privileges as a maintainer of the `vivliostyle/themes` repository (<https://github.com/vivliostyle/themes>):

- Push to the repository
- Triage issues
- Review and merge pull requests

Vivliostyle Themes Gallery

This is a gallery of Vivliostyle Themes released as npm packages. Newly released themes are added here as soon as we find them, but if you find something that is not listed, please send an [issue \(https://github.com/vivliostyle/themes/issues\)](https://github.com/vivliostyle/themes/issues) or pull request.

Official Vivliostyle Themes

Currently, there are six official themes.

@vivliostyle/theme-bunko (<https://www.npmjs.com/package/@vivliostyle/theme-bunko>)

It is suitable for Japanese vertical novels. It also supports ruby and horizontal-in-vertical composition (in Japanese, "縦中横"). CSS variables allow customization of lines/characters per page, page headers, and more.

- [README \(CSS variables list\) \(https://github.com/vivliostyle/themes/tree/main/packages/@vivliostyle/theme-bunko#readme\)](https://github.com/vivliostyle/themes/tree/main/packages/@vivliostyle/theme-bunko#readme)



Theme-bunko sample image. The manuscript is 'Night on the Galactic Railroad'.

@viviostyle/theme-slide (<https://www.npmjs.com/package/@viviostyle/theme-slide>)

This is good for slide presentations. The style changes between the cover page and the general page. CSS variables allow customization of colors, layout, and cover page styles.

- [README \(CSS variables list\)](https://github.com/viviostyle/themes/tree/main/packages/@viviostyle/theme-slide#readme) (<https://github.com/viviostyle/themes/tree/main/packages/@viviostyle/theme-slide#readme>)

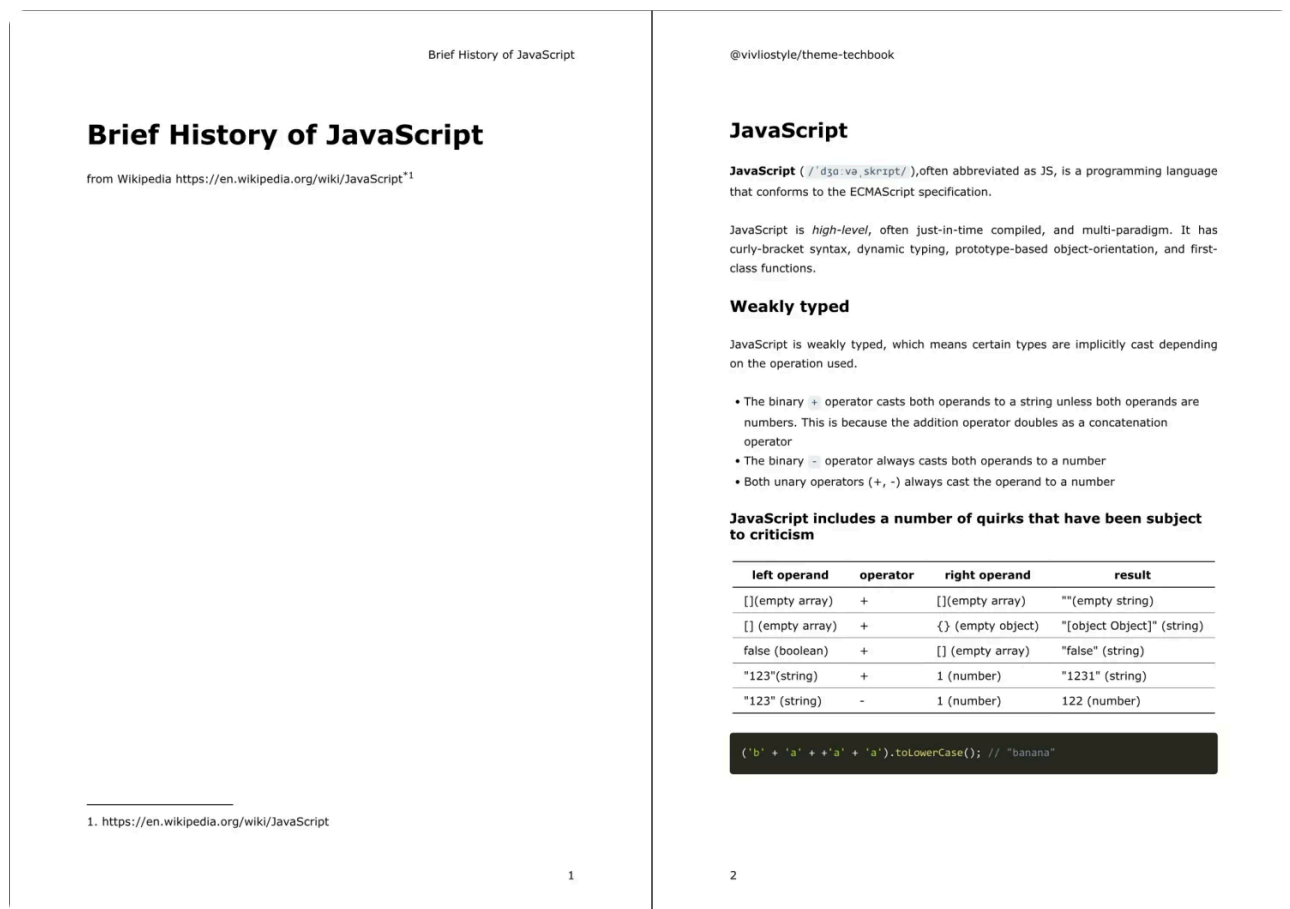


Theme-slide sample image. On the left is the cover page with a blue background, and on the right is the general page with a white background.

@vivliostyle/theme-techbook (<https://www.npmjs.com/package/@vivliostyle/theme-techbook>)

A technical book designed with printing in mind (e.g., margins adjustment at the inside/outside margin). Source code and table of contents are also supported. CSS variables allow customization of colors, page headers/footers, and image resolution.

- README (CSS variables list) (<https://github.com/vivliostyle/themes/tree/main/packages/@vivliostyle/theme-techbook#readme>)



Sample images from theme-techbook. In addition to text, it has a table and source code.

@vivliostyle/theme-academic (<https://www.npmjs.com/package/@vivliostyle/theme-academic>)

This is suitable for reports written by students. Chapter and section numbers are automatically added. CSS variables allow customization of figure sizes and frame element styles.

- README (CSS variables list) (<https://github.com/vivliostyle/themes/tree/main/packages/@vivliostyle/theme-academic#readme>)

vivliostyle theme academic のサンプル

vivliostyle theme academic のサンプル

vivliostyle theme academic のサンプル

FET の特性

2023/09/10

びよりお 増たか

1 目的

FET の特性を調査するとともに、FET を用いた回路の作製に習熟することを目的とする。

2 原理

2.1 半導体

物質は、電流が流れる導体と電流が流れない絶縁体。ある条件によって電流が流れる半導体に分類できる。これらはバンドギャップの幅の違いで区別でき、バンドギャップが大きいと電子が移動できず電流が流れない。半導体は絶縁体に比べてバンドギャップが小さく、半導体に不純物を混ぜることで電子や空孔の流れを制御できる。

半導体はシリコンやゲルマニウムがよく使われるが、それらの結晶の性質は絶縁体に近く、電圧をかけても電流はほとんど流れない。しかしそこに電子を添加したリンなどの不純物をほんの少し加えるだけで、導体のような性質に変化する。この電子を余計にした不純物が含まれる半導体を n (negative) 型半導体。逆に電子の少ない空孔基などの不純物が入った半導体を p (positive) 型半導体と呼ぶ。p 型半導体では、空孔と呼ばれる電荷が足りないが電荷の穴のりの働きをし、その電子が動き回るように振舞う。この n 型と p 型を結合させることにより、電流の向きによって電流が流れたり流れなかったりする整流作用が現れる。

2.2 電界効果トランジスタの動作原理

p 型半導体と n 型半導体を接合することを pn 接合という。pn 接合すると、p 型半導体と n 型半導体の境界部分で電子と正孔が結合し、境界部分の n 側に p、p 側に n の電気を帯びた領域ができる。この 2 つの領域は変遷層と呼ばれる。電流が流れない領域である。しかし、p 側から n 側に電圧をかけると、変遷層で結合できなかった電子と正孔が解放し、一時的に変遷層が小さくなる。変遷層の厚さが十分に小さくなった時、電子と正孔がそれぞれ変遷層を越えられようになると、多数電流が流れる。この p 側から n 側に電流が流れる方向を電流方向という。逆方向に電圧をかけると変遷層は大きくなり、多数電流は流れない。このように、pn 接合によって電流を行う素子をダイオードと呼ぶ。

FET は 3 本の端子が取り出し、それぞれゲート (G)、ドレイン (D)、ソース (S) と名前がついている。FET は n 型半導体と p 型半導体の組み合わせでできており、チャネルの物質的性質によって n チャネル型 FET と p チャネル型 FET に分類できる。n チャネル型 FET はドレインとソースの電極領域が n 型半導体で、その間が p 型半導体である。p チャネル型 FET はドレインとソースの電極領域が p 型半導体で、その間が n 型半導体である。図 1 に n チャネル型 FET、図 2 に p チャネル型 FET の構造を簡略化して示した。

図 3 静特性実験の回路

3.3.1 $V_{GS} - I_D$ 特性

$V_{DS} = 0\text{ V}$ から 4.5 V まで変化させたときの I_D の値を測定した。 V_{DS} が 3 V 、 10 V の場合のそれぞれについて行った。

3.3.2 $V_{GS} - I_D$ 特性

$V_{DS} = 0\text{ V}$ から 18 V まで変化させたときの I_D の値を測定した。 V_{GS} が 0.2 V 、 0.4 V 、 0.6 V 、 0.8 V の場合のそれぞれについて行った。

3.4 動特性実験

図 4 に示す回路を作製した。

図 4 動特性実験の回路

3.4.3 30 V を印加したときのゲート電圧 V_{GS}

回路に入力電圧として 30 V 印加した時のソースに対するゲートの電圧 V_{GS} を調べた。

3.4.4 入力電圧 f に対する出力 v_{tp} の変化

入力電圧に $f = 1\text{ kHz}$ の数値を入力し、 v_{tp} (出力) が v_{in} (入力) に同じような変化を示すか調べた。ただし、 v_{in} は 50 mV から 3000 mV まで変化させた。

また、出力電圧が読みだす時の v_{tp} を調べた。

3.4.5 電圧増幅率 μ の測定数値特性

入力電圧を 1 V とし、周波数 f を 0.004 kHz から 3000 kHz まで変化させ、電圧増幅率 μ の増減数値特性を調べた。静特性・動特性の実験で使った回路の一例を示す。

表 1 実験器具				
名称	製造会社	型番	製造番号	規格
FET	-	K31AG (RSD)	-	-
コンデンサ	-	80P4, 7414	-	10V, 16 S, max 5 F
直流電源	METRONIX	543A	178541	40V, 0.5A
デジタルマルチメータ	TEXIO	DL-2040	13010300	10V, 50/500Hz
オシロスコープ	IZO	IO-8020	80755246	200MHz
アンプ/ジャンジェネレータ	WATSON	W6-4105	3227463	150MHz
電圧計	KENWOOD	AG-2040	11070316	50/500Hz, 10.8V
テスタ	YOKOKAWA	T32-01	9861132	-

4 実験結果

4.5 静特性実験

4.5.6 $V_{GS} - I_D$

実験結果を図 5 に示す。 V_{GS} が 2 V 以下で I_D は V_{GS} が大きくなるにつれて減少し、 V_{GS} が 2 V 以上では I_D はほぼ 0 A となった。また、 V_{GS} の値が大きければ I_D の値は大きくなった。

4.5.7 $V_{GS} - I_D$ 特性

実験結果を図 6 に示す。 V_{GS} が 0 V から 2 V までの範囲で I_D の値は急激に増加し、 2 V 以降は増加減少を繰り返しながら全体的に増加した。 V_{GS} の値が大きければ I_D の値は小さくなった。

4.6 動特性実験

4.6.8 30V を印加した時のゲート電圧 V_{GS}

V_{GS} の値は 1.42 V であった。

1

3

4

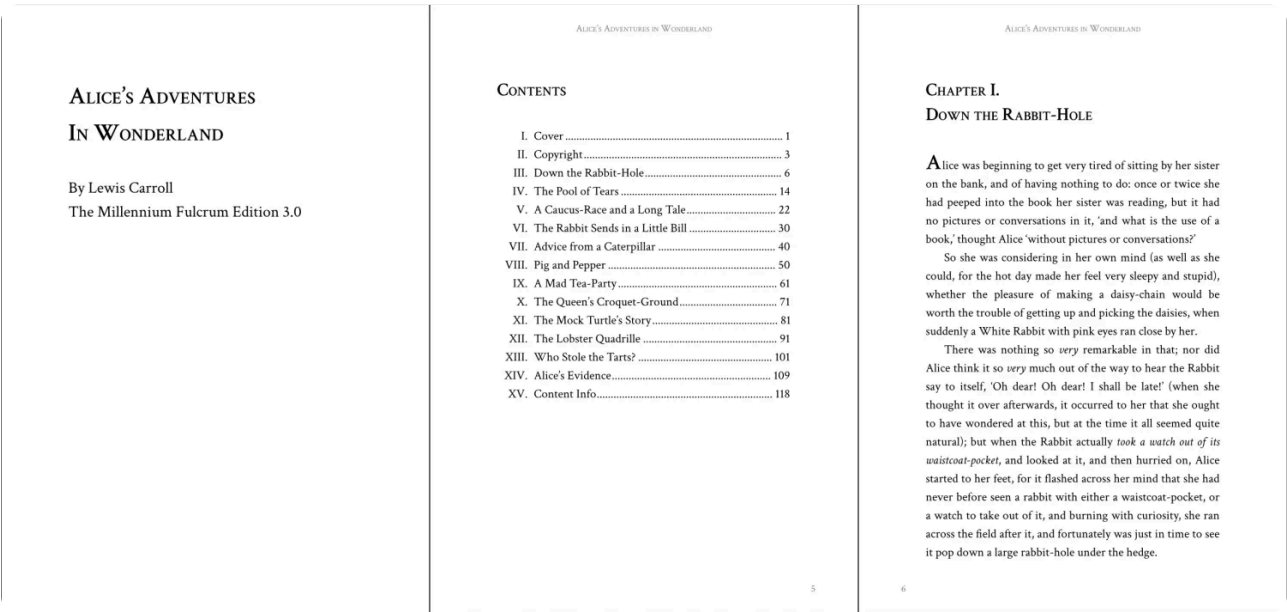
Sample images from theme-academic. In addition to text, it contains diagrams and tables.

@vivliostyle/theme-gutenberg (<https://www.npmjs.com/package/@vivliostyle/theme-gutenberg>)

This is suitable for horizontal novels in English. In addition to the default `theme.css`, three CSS variations (`alice.css`, `fang.css`, `sherlock.css`) are available.

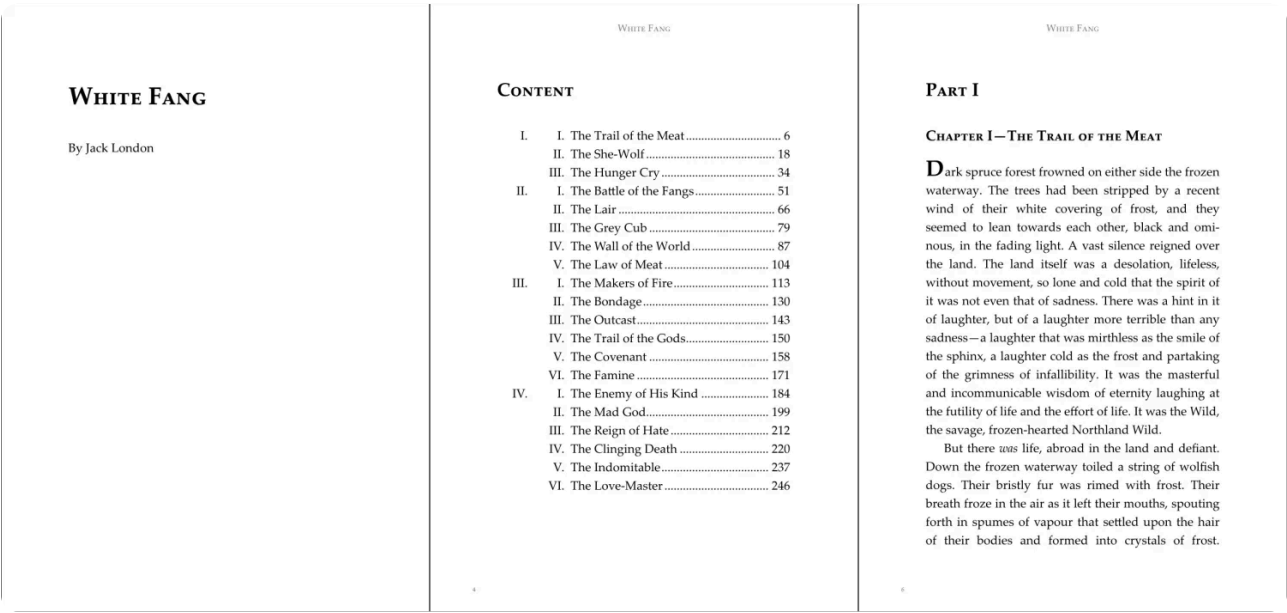
- README (<https://github.com/vivliostyle/themes/tree/main/packages/@vivliostyle/theme-gutenberg#readme>)

alice.css



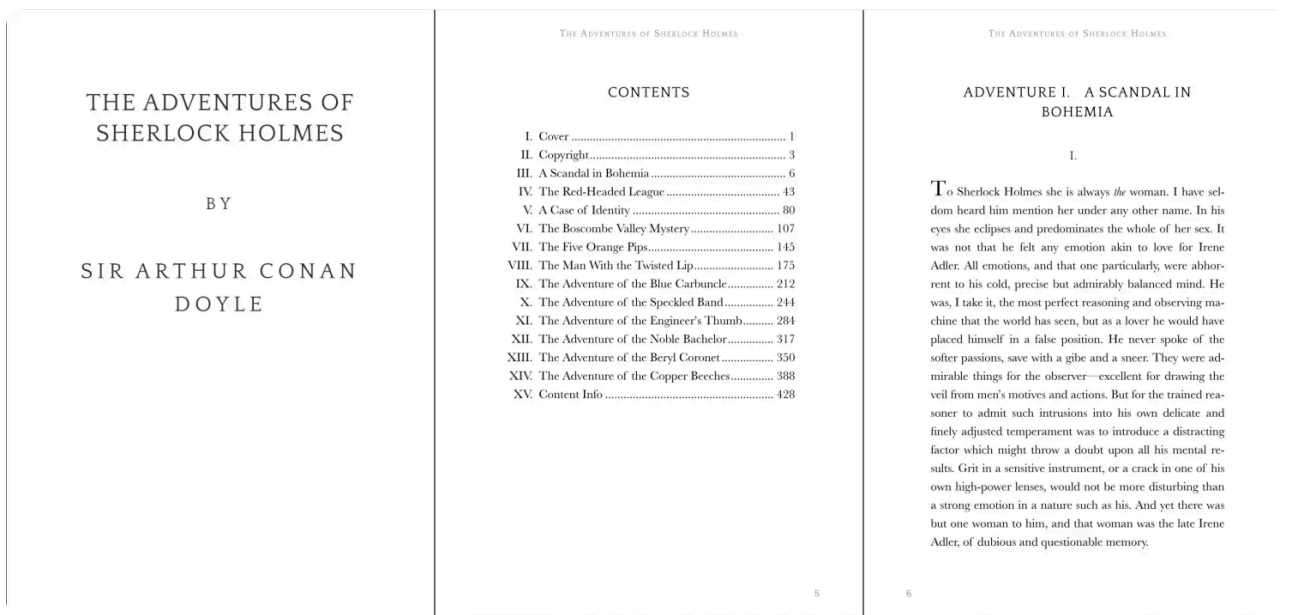
Sample image from theme-gutenberg. The manuscript is 'Alice's Adventures in Wonderland'.

fang.css



Sample image from theme-gutenberg. The manuscript is 'White Fang'.

sherlock.css



Sample image from theme-gutenberg. The manuscript is 'The Adventures of Sherlock Holmes'.

@vivliostyle/theme-epub3j (<https://www.npmjs.com/package/@vivliostyle/theme-epub3j>)

This is a theme for creating EPUB compatible with the EBPAJ EPUB 3 File Creation Guide (<http://ebpaj.jp/counsel/guide>). Suitable for Japanese vertical text EPUB publications.

- README (<https://github.com/vivliostyle/themes/tree/main/packages/@vivliostyle/theme-epub3j#readme>)



Screenshot of theme-epub3j example.

Unofficial Vivliostyle Themes

vivliostyle-theme dnd-5e-phb (<https://www.npmjs.com/package/vivliostyle-theme-dnd-5e-phb>)

Author: oldgeese

D&D 5e PHB theme for Vivliostyle

VIVLIOSTYLE THEME

D&D 5E PHB

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

HEADER2 ヘッダー2

Lorem ipsum dolor sit amet, consectetur adipiscing elit,

HEADER3 ヘッダー3

Lorem ipsum dolor sit amet, consectetur adipiscing elit,

HEADER4 ヘッダー4

Lorem ipsum dolor sit amet, consectetur adipiscing elit,

HEADER5 ヘッダー5

Lorem ipsum dolor sit amet, consectetur adipiscing elit,

Header6 ヘッダー6

strong, 太文字,

italic, イタリック,

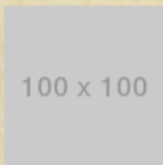
親譲りの無鉄砲で小供の時から損ばかりしている。小学校に居る時分学校の二階から飛び降りて一週間ほど腰を抜かした事がある。なぜそんな無闇をしたと聞く人があるかも知れぬ。別段深い理由でもない。新築の二階から首を出していたら、同級生の一人が冗談に、いくら威張っても、そこから飛び降りる事は出来まい。弱虫やーい。と囃したからである。小使に負ぶさって帰って来た時、おやじが大きな眼をして二階ぐらいから飛び降りて腰を抜かす奴があるかと云ったから、この次は抜かさずに飛んで見せますと答えた。(青空文庫より)

UNORDERED LIST

- **a horizontal rule before a ul** will give it the style of unordered list without a bullet.

a horizontal rule before a ul will give it the style of unordered list without a bullet.

IMAGES



placeholder

CLASS A FEATURES

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

TIME TO DROP KNOWLEDGE

Use notes to point out some interesting information.

Tables and lists both work within a note.

TIME TO DROP KNOWLEDGE

Use notes to point out some interesting information.

Tables and lists both work within a note.

THE NOTARY

Level	Proficiency Bonus	Features	Sixgun Poisoner
1st	+2	Gunpowder Torturer	+1
2nd	+2	Pharmaceutical Gunslinger	+2
3rd	+2	Gunslinger Corruptor	+2
4th	+2	Ability Score Improvement, Demonic Anthropology	+3
5th	+3	Torque Interfacer	+4
6th	+3	Ability Score Improvement, Mathematical Pharmacist	+4
7th	+3	Sixgun Poisoner	+4
8th	+3	Ability Score Improvement, Spiritual Biochemistry	+4
9th	+4	—	+4
10th	+4	Consecrated Biochemistry	+4
11th	+4	Torque Interfacer	+4
12th	+4	Ability Score Improvement, Ritual Astronomy	+4
13th	+5	Plasma Outlaw	+4
14th	+5	Ability Score Improvement, Plasma Outlaw	+5
15th	+5	Civil Alchemy	+6
16th	+5	Ability Score Improvement, Spell Analyst	+6
17th	+6	—	+6
18th	+6	Exo Interfacer	+6
19th	+6	Ability Score Improvement, Police Cultist	+6
20th	+6	Nuclear Illusionism	+6

HIT POINTS

Hit Dice: 1d12 per level

vivliostyle-theme-thesis (<https://www.npmjs.com/package/vivliostyle-theme-thesis>)

Author: yamasy1549

Academic theme (thesis)

vivliostyle-akashi-kosen-bulletin (<https://www.npmjs.com/package/vivliostyle-akashi-kosen-bulletin>)

Created by: yamasy1549

A style for annual report of National Institute of Technology, Akashi College (unofficial)

Spec of Vivliostyle Theme

If you want to publish your theme as an npm package, please follow the specification below. You can use `create-vivliostyle-theme` (<https://github.com/vivliostyle/themes/tree/master/packages/create-vivliostyle-theme>) to quickly create a theme that follows this specification.

Theme Name

The name of the theme can be free. However, it is recommended to follow some points:

- If the purpose is clear, include a concise term that represents it.
- Avoid terms related to modifiable styles.
 - For example, publication size and layout direction are expected to be easily changeable by users in the future with tools like Vivliostyle Pub. Therefore, avoid using terms related to these modifiable styles.
 - Examples of names to avoid: `vivliostyle-theme-a4book`, `vivliostyle-theme-tategaki`, etc.

Styles to Include in the Theme

A theme should encompass styles for the entire publication, rather than focusing on specific elements (e.g., only figures or footnotes).

Below are the typical styles that should be included in a theme. The theme template generated with `create-vivliostyle-theme` (<https://github.com/vivliostyle/themes/tree/master/packages/create-vivliostyle-theme>) includes these styles.

- Page (margin, running header/footer, etc.)
- Heading
- Paragraph
- Math expression
- Figure and caption
- Table and caption
- Source code
- Footnote

- Ruby

Directory Structure

```
vivliostyle-theme-mybook/
├─ LICENSE
├─ README.md
├─ package.json
├─ example/
│   └─ ...
│       └─ default.md
├─ theme.css
└─ vivliostyle.config.js
```

When you create a theme template using `create-vivliostyle-theme` (<https://github.com/vivliostyle/themes/tree/master/packages/create-vivliostyle-theme>), the essential files mentioned above will be generated automatically. The details of some typical files are described in the following sections.

package.json

```
{
  "name": "vivliostyle-theme-mybook",
  "author": "John Doe <john@example.com>",
  "main": "theme.css",
  "keywords": [
    // described below
    "vivliostyle",
    "vivliostyle-theme"
  ],
  "vivliostyle": {
    // described below
    "theme": {
      "style": "./theme.css",
      "name": "Mybook",
      "author": "John Doe",
      "category": "novel",
      "topics": ["paperback"]
    }
  }
}
```

You can use `vivliostyle-theme-scripts` (<https://github.com/vivliostyle/themes/tree/master/packages/vivliostyle-theme-scripts>) to check that the `package.json` of your theme conforms to the specification.

```
$ vivliostyle-theme validate
```

keywords Property

Optional.

Including `"vivliostyle-theme"` in this property's value will list your theme among the available options when creating a publication using `Create Book` (<https://github.com/vivliostyle/create-book>). Ensure your theme is published as an npm package for it to appear.

Since a theme is an npm package related to Vivliostyle, it is advisable to include `"vivliostyle"`.

vivliostyle.theme Property

style Property

Required. This property specifies the main CSS file in the theme.

You can also use the `style` or `main` properties at the top level of `package.json`. These properties serve the same purpose as `vivliostyle.theme.style`, but the priority order is `vivliostyle.theme.style > style > main`.

```
{
  "main": "theme.css",
  "style": "theme.css"
}
```

author Property

Required.

category Property

Optional. This property provides a hint to users about the primary use of your theme when they use it for the first time.

Choose the category that best fits your theme from the following list:

- "novel"
- "magazine"
- "journal"
- "report"
- "misc"

Note that this list may be updated in the future.

topics Property

Optional. If you want more specific descriptions of the theme's use than the `category` property, you can list and describe them here.

example/

The example directory should contain at least one Markdown file, a straightforward example of a theme. The name of the file is arbitrary.

In Markdown files, you can use VFM (Vivliostyle Flavored Markdown) (<https://vivliostyle.github.io/vfm/#/vfm>).

theme.css

Include the stylesheet that defines the theme's styles. Using CSS custom properties (CSS variables) is recommended so that users can flexibly customize the theme's styles.

A theme can include multiple stylesheets. For example, besides the default `theme.css`, you can provide stylesheets for specific purposes as shown below:

- `theme_print.css` : Defines styles for outputting as PDF or viewing with Vivliostyle Viewer.
- `theme_screen.css` : Defines styles for outputting in webpub format such as HTML.

To specify a stylesheet other than the default with Vivliostyle CLI, set the `specifier` and `import` properties as follows:


```
theme: {
  specifier: 'vivliostyle-theme-mybook',
  import: 'theme_print.css',
},
```

vivliostyle.config.js

```
/** @type {import('@vivliostyle/cli').VivliostyleConfigSchema} */
module.exports = {
  language: 'en',
  theme: ['node_modules/@vivliostyle/theme-base', '.'],
  entry: ['example/default.md'],
  workspaceDir: '.vivliostyle',
  output: [
    'book.pdf',
    {
      path: './book',
      format: 'webpub',
    },
  ],
};
```

This is a configuration file for generating publications from Markdown files in example. It is used by developers to develop themes and by theme's users to generate sample publications.